

Intro to Laser Engraving

Jon A 5/2/26



Topics Covered

- Machine Basics
- Choose and customize a design
- Import into Lightburn for cutting
- We will set up cutter for operation
- You will laser etch your own cutting board to take home

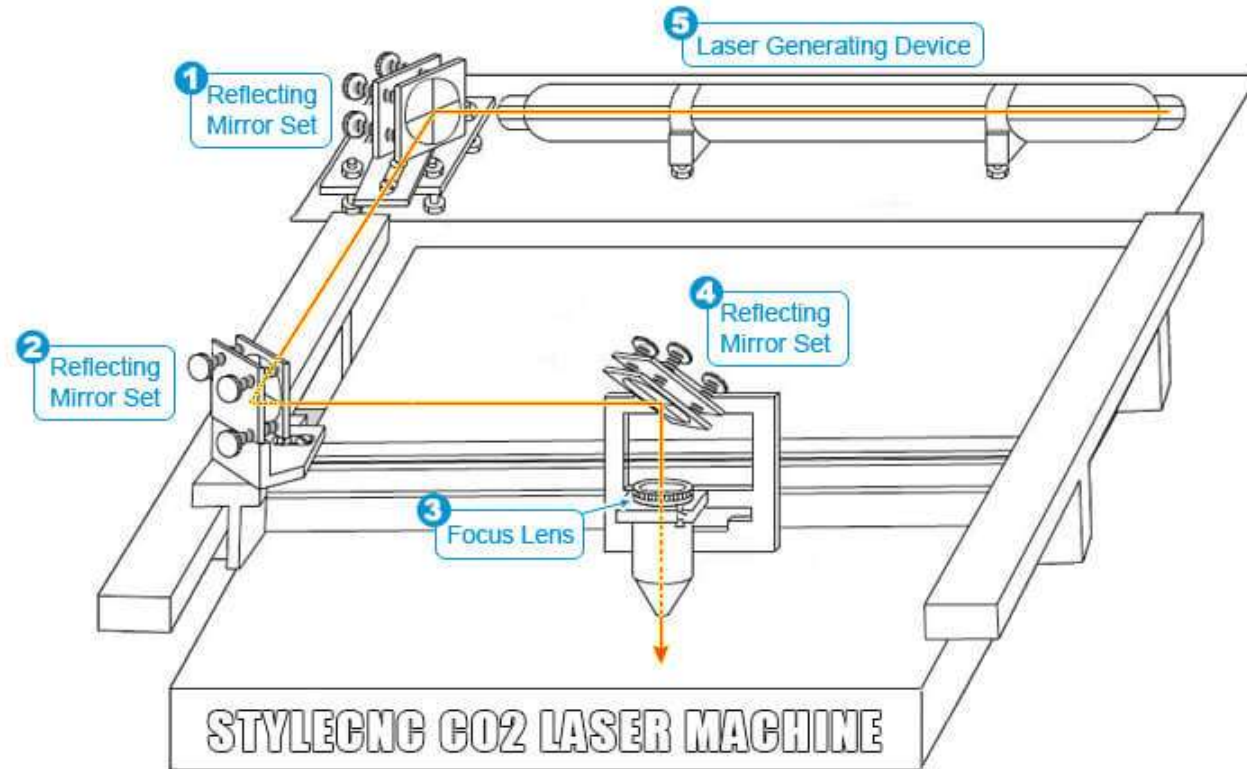


Machine Basics

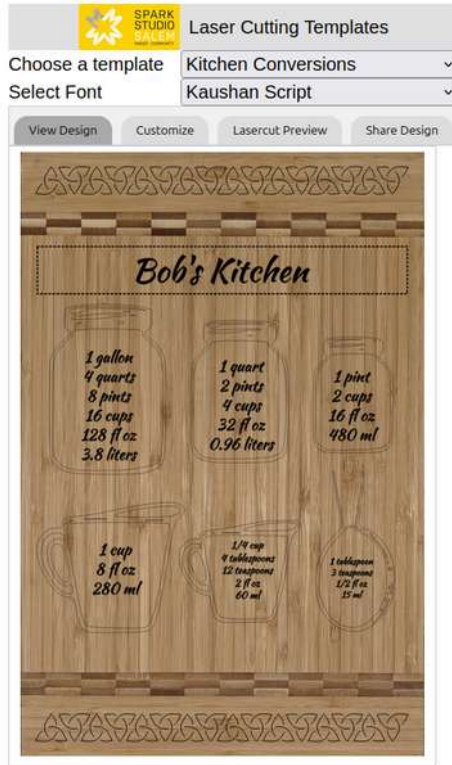
- 150 Watt CO2 Laser (10.6um wavelength)
- Water cooled
- Adjustable bed height
- Usable Work Area: 550mm x 850mm (21.5in by 33.4in)
- Air extraction
- Air assist to improve cut quality/move debris away from nozzle



Machine Basics



Choose a template



Open template tool

- Select and customize file for cutting
- Download SVG file to save it
- Upload SVG file to cutting file server at:
- Lightburn for sending to laser



Advanced Designs

- Inkscape is a great resource for more advanced designs
 - Free and open-source
- Other packages like Canva or Adobe Illustrator are suitable.
 - More expensive, but sometimes easier to use
- Great YouTube resources
- Check Handout for more information

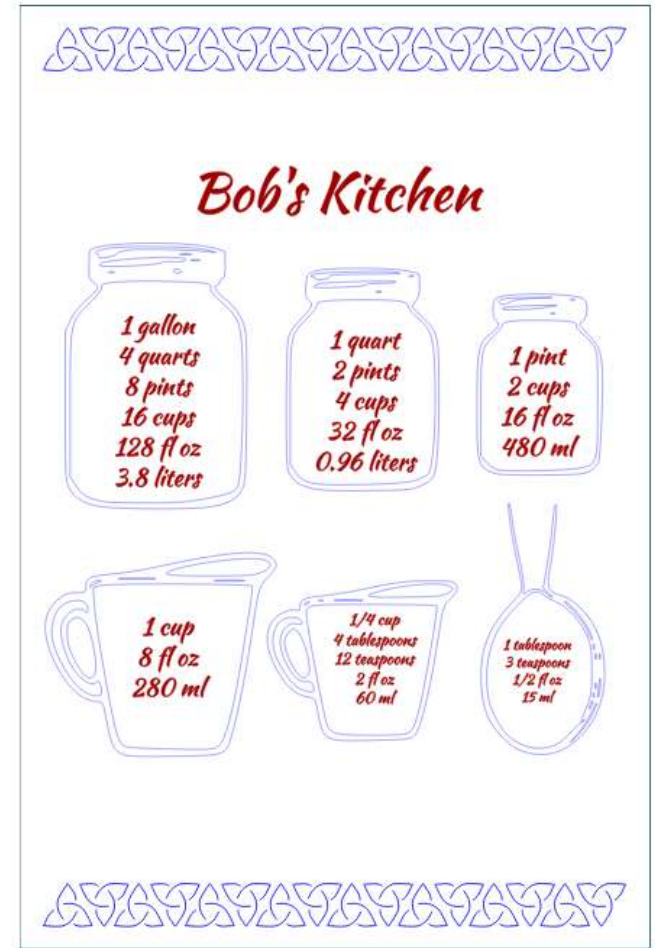


Finished Design

Design starts at 60mm, 60mm

Fits within the cutting board shape

Colors will be used to plan what type of operation the laser will perform.



Import your design

- Go to Lightburn PC
- Download your SVG file
- Align design at $x=60\text{mm}$, $y=60\text{mm}$
- Matches physical location on machine



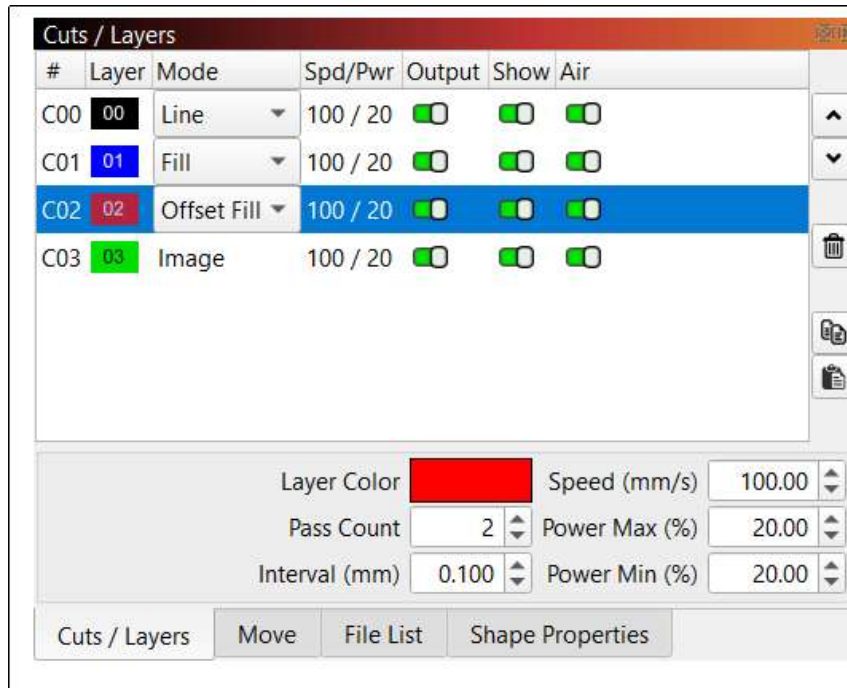
Setting Feeds and Speeds

- Each element in your design should be given a “color”. That “color” will later determine how the cuts happen.
- Each color gets a “speed”, “power” and “mode”.
- Speed/Power determine how “deep” the laser will cut
- Mode determines how the cut is made



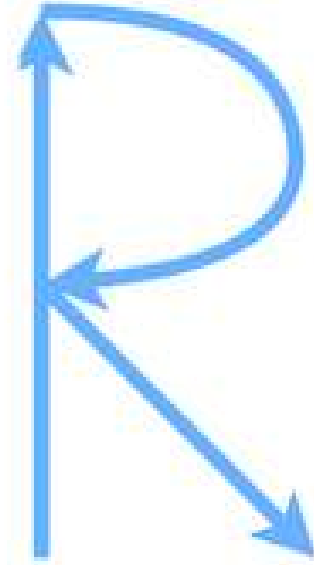
Setting Feeds and Speeds

- Each “color” gets a speed and power, and mode
- Never exceed 50% power



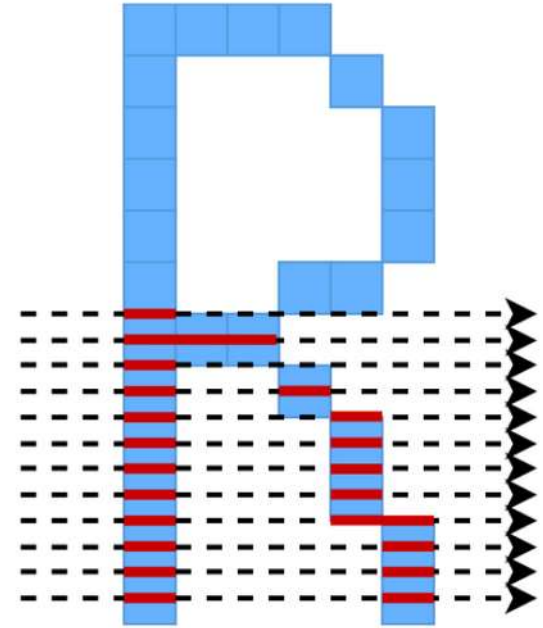
Modes

- Line: Simply follows the lines in a single pass
- Very thin lines sweeping across the design
- Good for line art and cutting shapes



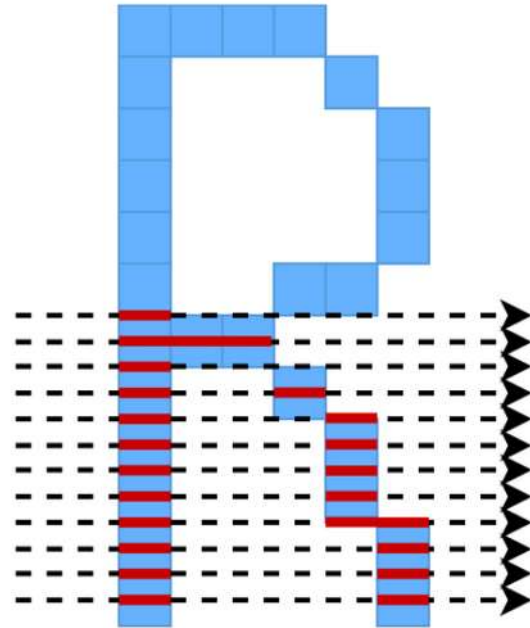
Mode: Image

- Scans across the image line by line
- Pulses beam on and off to mark surface



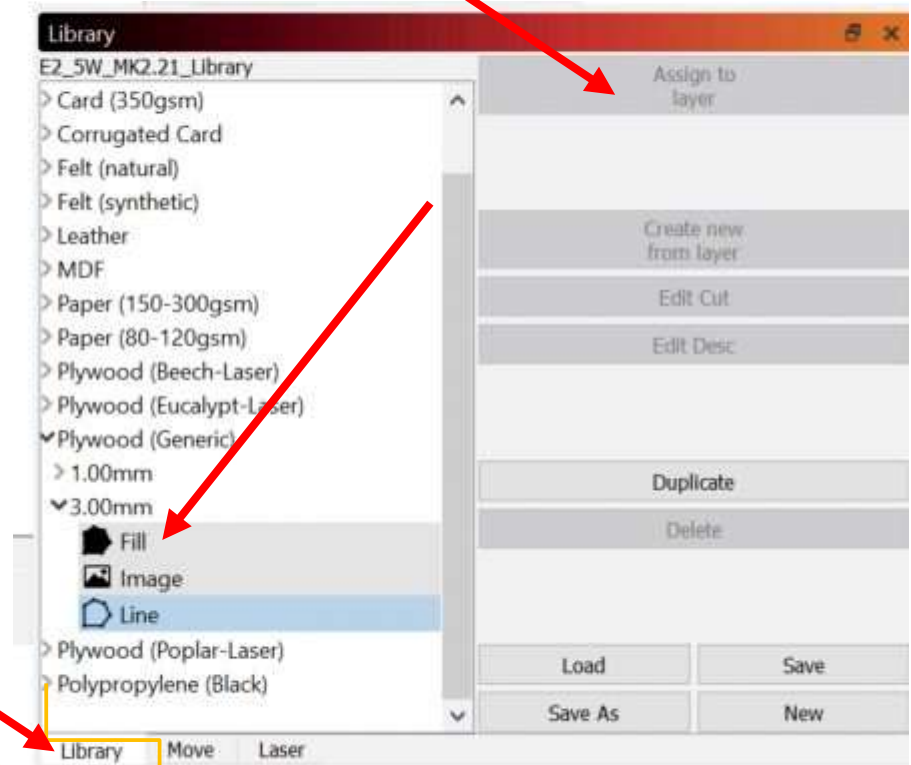
Mode: Fill, Offset Fill

- Same as image, but used to fill in regions
- Offset fill doesn't scan line by line, but instead repeatedly traces lines in concentric patterns to fill



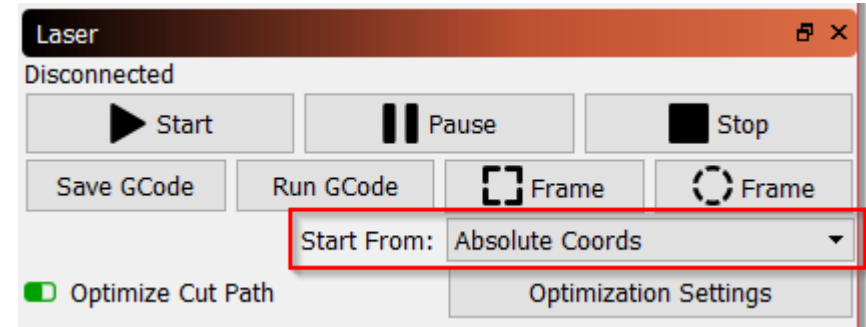
Material Library

- Assign layers (colors).
- Each one is a pre-set speed, power, and mode.
- Allows us to quickly determine how it will cut.
- You can still “fine-tune” speed and power afterwards if you want.



Send to machine for cutting

- Once speed and power are set, we can send our design to the machine to cut it.
- SEND button
- Set “Start From” to “Absolute” coordinates



Making the cuts

- Put material in machine
- Set focal distance between work and laser head to 5mm
- Close all doors/panels.
- Select File
- Press “Frame” button to confirm position
- Press green “Start/Pause” button.
- Don’t stare at beam
- Don’t leave unattended



Materials

- Materials: Approved

- MDF
- All wood (except pressure-treated lumber)
- Acrylic (Plexiglass)
- UNTREATED or Vegan treated leather
- Stone, Glass, Ceramic
- Metal for surface engraving only

- Materials: Banned

- Plastics containing chlorine
 - Vinyl, PVC, Nylon
- Resins
- Epoxies



Safety

- Make sure all panels are closed
- Beware of fire, wood can burn.
- Beware in case lens vaporizes, possible Selenium exposure
- Do not touch mirrors or nozzle
- Persons under 18 must have parent or guardian nearby



Next Steps

- After this training, feel free to come to the space during open-hours and use the machines provided that an authorized trainer is available to supervise machine operation. You will have to provide your own designs and materials, but use of the machines is free.
- If you become a member, there is additional training to:
 - Operate unsupervised or during non open-house hours.
 - Maintain and align machine
- If you enjoy this, we encourage you to become a member and get actively involved in the community.
- Tell your friends how much fun it is!



Now, let's go do some cutting!

- We will now move into the main space and make some cutting boards.
- Have fun with it and be creative.



Next slides are just for reference



Requirements

- Must be supervised by authorized user
- Must use only during office hours or open-house.
- 18+ or supervised by parent/guardian
- Donation to Spark
 - \$40 Non-members
 - \$20 Members
 - Please help us, it takes time and effort to keep these machines in good condition.
 - We also have to pay for facilities, electricity, etc.
 - You will not find a better deal anywhere!



Trace bitmap to path

- OPTIONAL
- You can “convert” the image into a set of paths for better control
- Somewhat tricky to get the right effect, but can get really good results.
- Once it's a path, you can edit nodes for very fine-grained control
- Works best when there's a lot of contrast in the image

